

PROJECT PROPOSAL

Digital Grade-Report Locker (DGRL)

A Secure Mobile-Based Academic Result Management System

1. Introduction

With the rapid growth of digital technology in education, universities are shifting from manual systems to automated solutions. However, many institutions still rely on traditional methods such as notice boards or limited portals to distribute student results. These approaches are inefficient, insecure, and inconvenient.

The **Digital Grade-Report Locker (DGRL)** is a mobile-based solution developed using **Flutter** to provide students with secure, fast, and reliable access to their academic results anytime and anywhere.

2. Problem Statement

The current academic result systems face several challenges:

- **Privacy Issues:** Student results displayed publicly expose sensitive information.
- **Limited Accessibility:** Students cannot easily access results remotely.
- **Manual Processes:** GPA/CGPA calculations are often done manually.
- **Poor Record Management:** Difficulty in retrieving past academic records.
- **Lack of Mobile Support:** Most systems are not optimized for smartphones.

3. Objectives

3.1 General Objective

To design and develop a mobile application that securely manages and displays student academic results.

3.2 Specific Objectives

- Develop a secure authentication system for students
- Provide real-time access to grades
- Automatically calculate GPA and CGPA
- Maintain historical academic records
- Enable offline access using local storage
- Generate downloadable PDF reports

4. Scope of the Project

4.1 Included Scope

- Student login and authentication
- Viewing grades per semester
- GPA and CGPA calculation
- Result history tracking
- PDF report generation
- Basic admin functionalities

4.2 Excluded Scope

- Full university management system
- Course registration system
- Payment integration

- Advanced analytics system

5. System Overview

The system consists of two main components:

1. Mobile Application (Frontend)

- Built using Flutter
- User-friendly interface
- Cross-platform (Android & iOS)

2. Backend System

- Firebase Firestore for database
- Firebase Authentication for login
- Cloud-based data storage

6. Technical Specifications

Layer	Technology
Frontend	Flutter
Programming Language	Dart
Backend	Firebase
Database	Cloud Firestore (NoSQL)
Authentication	Firebase Auth
Local Storage	Hive / SharedPreferences
Development Tools	Android Studio / VS Code

7. Functional Requirements

7.1 Student Functions

- Login using Student ID and password
- View dashboard with GPA summary
- Access semester-wise results
- View detailed grade breakdown
- Download results as PDF
- Logout securely

7.2 Admin Functions

- Add student records
- Add courses
- Upload and update grades
- Manage academic data

8. Non-Functional Requirements

- **Security:** Data encryption and secure authentication
- **Usability:** Simple and intuitive interface
- **Performance:** Fast loading and response time
- **Reliability:** Minimal downtime using cloud services
- **Scalability:** Can support many users

9. Database Design

Entities and Attributes

Student

- student_id
- full_name
- department
- batch
- password

Course

- course_code
- course_title
- credit_hours
- instructor

Grade

- student_id
- course_code
- mid_score
- assignment_score
- final_score
- total_score
- letter_grade

10. System Architecture

The system follows a **client-server architecture**:

- **Client:** Flutter Mobile App

- **Server:** Firebase Backend
- **Database:** Firestore

Data flows from the mobile app to Firebase and is retrieved in real-time.

11. User Interface Design (Screens)

- Splash Screen
- Login Screen
- Dashboard
- Semester Results Page
- GPA Calculator
- Profile Page
- PDF Export Page
- Admin Panel

12. Methodology

The project will follow the **Agile Development Model**:

مراحل (Phases):

1. Requirement Analysis
2. System Design
3. Implementation (Flutter + Firebase)
4. Testing
5. Deployment
6. Documentation

13. Expected Results

- Secure and private grade access
- Faster result checking
- Accurate GPA/CGPA calculation
- Improved student experience
- Reduced paper-based processes

14. Advantages of the System

- Mobile accessibility
- Real-time updates
- Easy to use
- Scalable system
- Cost-effective

15. Limitations

- Requires internet connection (initially)
- Depends on Firebase availability
- Limited admin functionality (basic version)

16. Future Enhancements

- Push notifications for new results
- Attendance tracking
- Course registration module
- Chat system with instructors

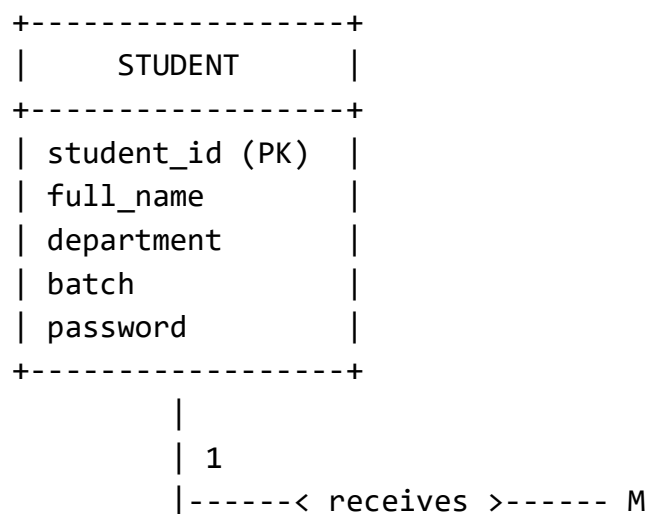
- Multi-language support

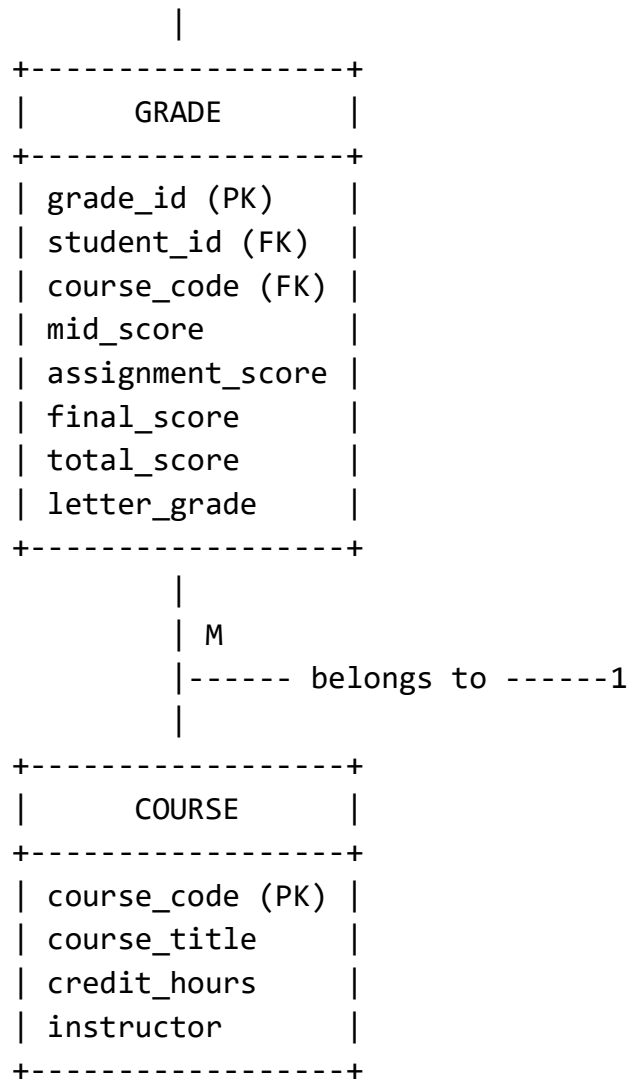
17. Project Significance (Very Important for Grading)

- Solves a real-world university problem
- Demonstrates mobile development skills
- Uses modern technologies (Flutter + Firebase)
- Shows database design understanding
- Easy to implement within course timeline

18. Diagrams

18.1 Entity Relationship Diagram (ERD)

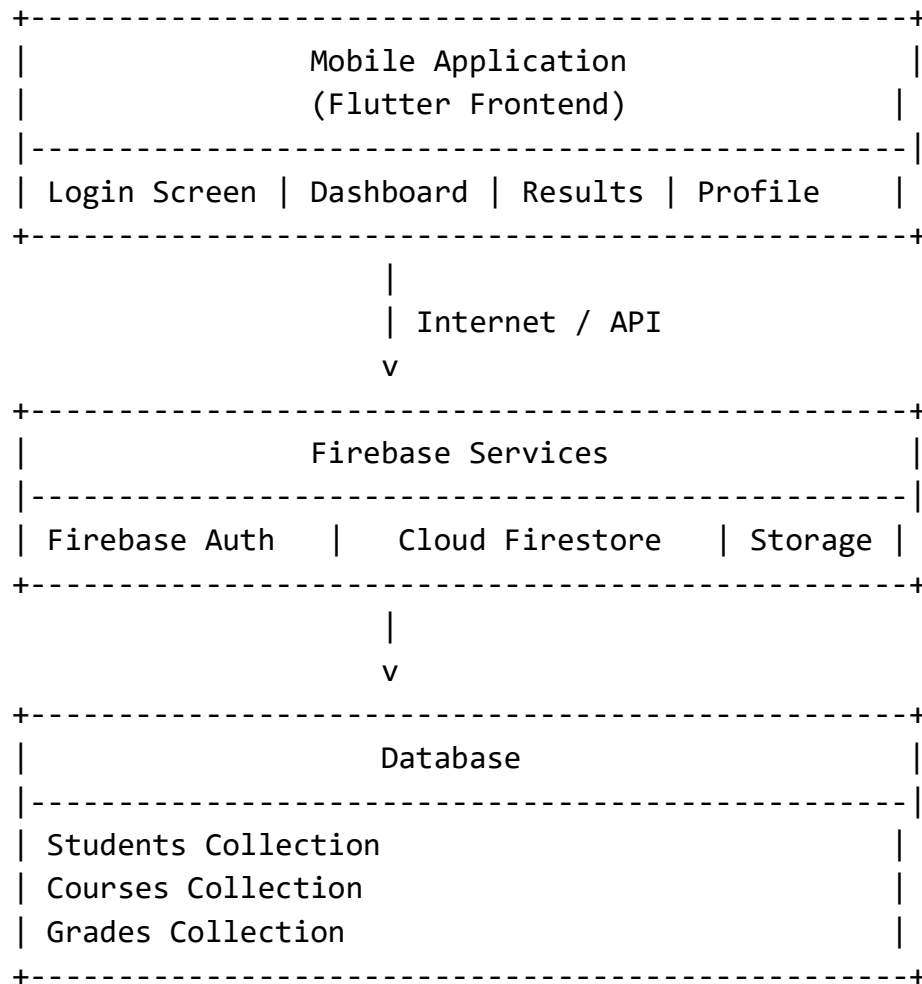




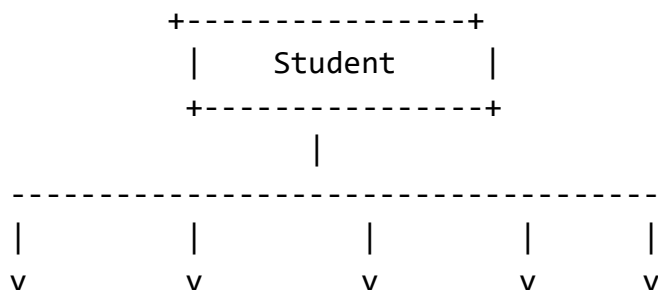
Relationship Explanation:

- One **Student** can have many **Grades**
- One **Course** can have many **Grades**
- Grade connects Student and Course

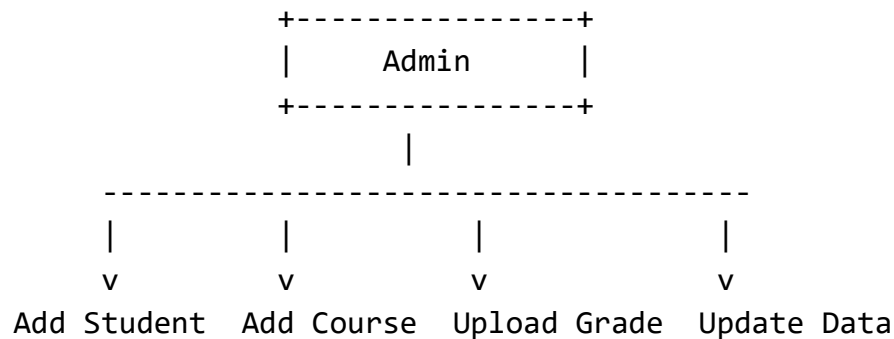
18.2 System Architecture Diagram



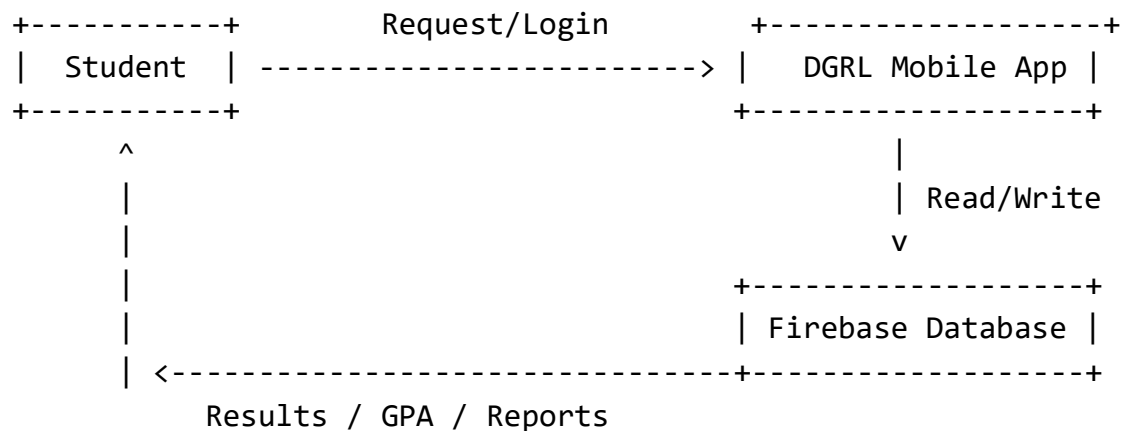
18.3 Use Case Diagram



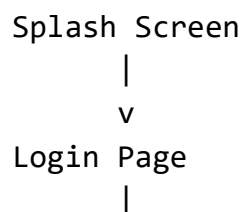
Login View Grades GPA Calc PDF Logout

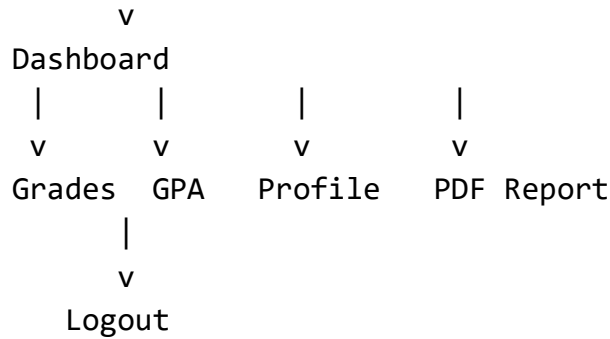


18.4 Data Flow Diagram (DFD - Level 0)

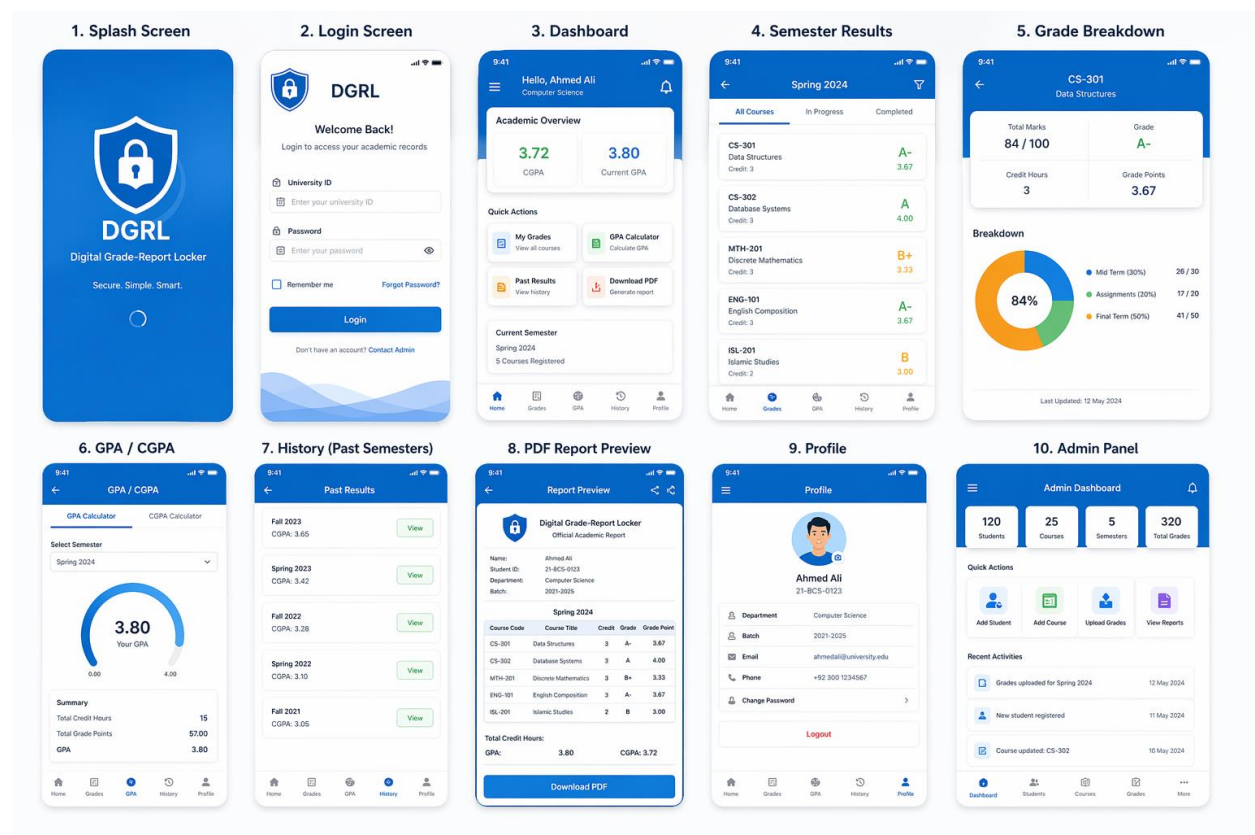


18.5 Navigation Flow Diagram





18.6 Flutter UI screenshots design



19. Conclusion

The Digital Grade-Report Locker provides a modern solution to traditional academic result systems. By leveraging Flutter and Firebase, the application ensures secure, efficient, and

user-friendly access to academic data. It is a practical and scalable project suitable for academic purposes and real-world implementation.